# The Ephemeris

**November 2014**
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## SOLAR ECLIPSE October 23, 2014

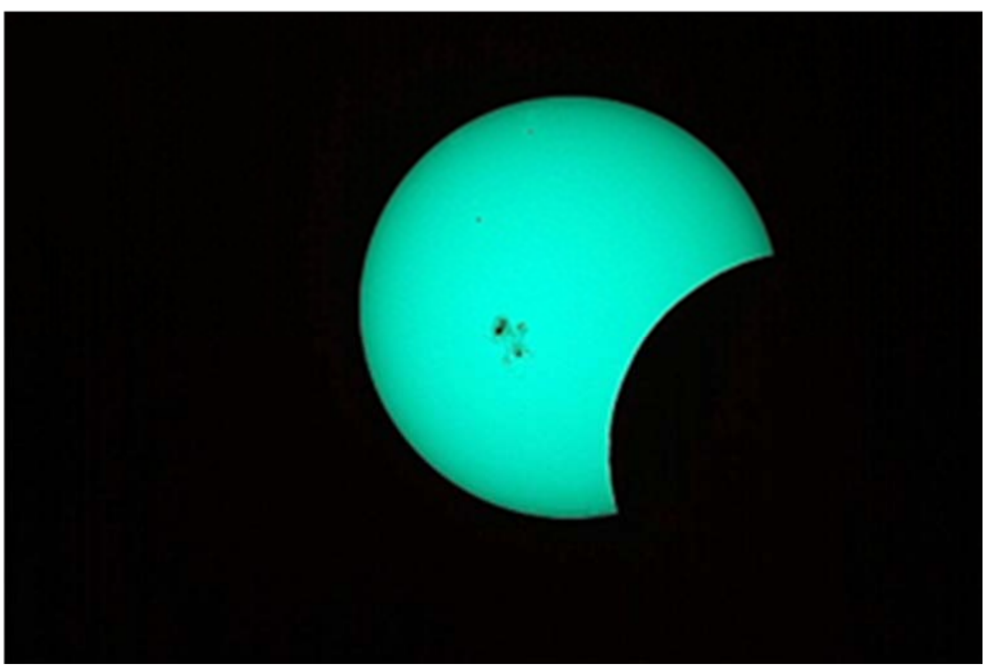
From SJAA President, Rob Jaworski

The San Jose Astronomical Association hosted the partial eclipse observing event at it's base camp; San Jose's Houge Park. The objective was to allow people to safely view the eclipse, share views with each other and the public, and to simply socialize.

The club's solar workhorse of a telescope, a 100mm hydrogen alpha scope made by Lunt, was doing its thing. A host of other telescopes were set up by club members and other people, many of which had filters attached for safe viewing, or projection systems installed so that the image could be viewed by many people at once, not unlike viewing the event on a small smartphone screen . Some people even brought out spaghetti strainers, the holes of which act as pin hole cameras, projecting numerous crescent sun images onto the ground, a white piece of paper, or even a hand.

Special solar viewing glasses, which make it safe for people to look directly at the sun, were made available to all attendees. Kids and adults alike were able to wrap these glasses across their eyes and have a safe, direct view at any time. And when we did, we were treated not just to a portion of the moon covering the sun, but there was also an additional treat: An unusually enormous sunspot was clearly visible to the naked eye. Everyone who knew what they were looking at could easily spot the aberration on the face of the sun. What made this special is that sunspots are typically much smaller, and not visible without some amount of magnification. The eclipse today was accentuated by this sunspot, which did not at any time become obscured by the moon's limb, and whose size is fairly rare.

During the roughly three and a half hours of the event, an estimated seventy five people attended. Some were seasoned amateur astronomers who were ready with their own equipment, while others were families that happened onto the event and were invited over by friendly volunteers of the SJAA. (continued on page 2)



*Picture: An unusually enormous sunspot which coincided with the eclipse*
*Photo Credit: Glenn Newell*

## November 2014 Events

**Sunday, Nov 02**
Solar observing: 2-4PM
Fix-It Day: 2-4PM

**Saturday, Nov 08**
Board Meeting: 6 -7:30pm
General Mtg: 7:30-10pm

**Sunday, Nov 09**
SJAA Annual Swap Meet: 12pm

**Friday, Nov 14**
Beginner Astronomy Class: 7pm
In-Town Star party (Houge): 7-9pm
Binocular Star Gazing (RCDO)

**Saturday, Nov 15**
Starry Nights Star Party (RCDO): 7-9pm

**Saturday, Nov 22**
Henry Coe—Dark Sky Weekend

**Friday, Nov 28**
In-Town Star party (Houge): 7-9pm

**Saturday, Dec 06**
Board Meeting: 6 -7:30pm
General Mtg: 7:30-10pm

**Sunday, Dec 07**
Solar observing: 2-4PM
Fix-It Day: 2-4PM

***SJAA events are subject to cancellation due to weather. Please visit website for up-to-date info.***

### INSIDE THIS ISSUE

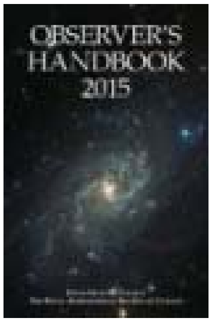
# SOLAR ECLIPSE October 23, 2014

*Solar Eclipse event at Hogue Park  hosted by SJAA*
*Photo credits: Sandy Mohan, Glenn Newell and Michael Packer*

(Continued from page 1)
Kids and retirees alike viewed the eclipse (and sunspot) in amazement, while everyone enjoyed the mild autumn weather of northern California.

This will be the last solar eclipse that will be visible in California for the next few years. In August of 2017, however, a large swath of North America, in particular the United States, will host a total eclipse come through. This will be a highly anticipated event, so now is the time to plan for seeing this total eclipse, which should be on everyone's list of thousand-things-to-do-before-you-die!

# Observer's Handbook 2015




The Royal Astronomical Society of Canada publishes each year its Observer's Handbook, which is a wonderful reference of astronomical data and phenomena of the year. Its target time period is mainly relevant for the year of publication, but much of its content is a wealth of information applicable to any time period of the epoch. The 2014 RASC Handbook noted four eclipses of the year, a total lunar eclipse in mid-April followed by an annular solar eclipse two weeks later (which was visible only in the far southern and eastern hemispheres), then another total lunar eclipse in early October and finally a partial solar eclipse on 23 October. This last solar eclipse was well situated for us observers in the western edge of North America.

# Binocular Astronomy - Why I Stopped Worrying and Learned to Love the Obie 25x100

From Manoj Kaushik

Many of you know me. I have been enjoying this hobby for a while now and have observed through most types of scopes. Until now, I was primarily using a 16" truss dob for enjoying the night sky. As you can imagine this takes some doing: Load the behemoth in pieces into your car, drive to Rancho or Mendoza. Spend 30 minutes setting it up. Then wait for it to cool down before collimating. Just listing all of them makes me tired. Of course there are rewards at the end of all this, but there's effort. Not for one of those nights when you are low energy.

Enter binoculars. The first time I got smitten by these was when Ed let me take a peek through his at Pinnacles. Being able to use both eyes, the wide FOV and the capability to pan effortlessly on a P-mount had me saying "this is how it should be done!"

The climb up the milky-way from the teapot all the way to Lagoon and Trifid had me mesmerized. Especially to be able to go between naked eye and the binos with ease without losing context and a sense of place in the overall night sky. By comparison, looking through the much narrower FOV of my 16" is like looking through a porthole. Detail and capability to hunt down tiny faint objects is fantastic, but context is lacking.

Well, by the time I had driven back home that night, I had decided I was going to get one. 2 or 3 weeks later I had my full rig on hand. I use a P-mount from universal astronomics and the setup of the 25x100 on the wood tripod and P-mount takes a barely longer than the time it takes to take them out of your car. Once setup is complete there is not much else to do. Just start observing! It's like microwave pop-corn: All the fun but none of the hassle.

There are down-sides, of course. You can't hunt those small fuzzies, nor those planetaries. But some nights, you don't want to. Some nights all you want is to pan the sky and just contemplate. You want them big nebulas and open clusters. Maybe the globs can join the party too. Now that I have the binos more of those nights have occurred. And I see a lot more of those nights happening in the future, especially when I travel!

All said, if you just want to learn the night sky, there is no better way to do it then a pair of good binos. Handhelds are great. But something like a 25x on a tripod is even better. Look through the bino, then look without. Now wonder where you are looking in the sky. This is probably the time when that smartphone becomes useful and starts teaching where the objects are relative to each other and in relation to you. Good way to get to know our friend, the night sky.

# Oct 8 Lunar Eclipse and More

by Mark McCarthy

When I checked the Clear Sky Chart for Fremont yesterday afternoon it predicted cloudiness, so I did not plan to observe the eclipse.  Then at 11:00pm before going to bed, I checked again on a whim and found it predicted clear skies & above average seeing & transparency.  So I rushed to move my scope (12.5" f/7 Dobsonian) from the garage to the far end of my back yard so I'd have good visibility to the west, where the moon would be at the time of the eclipse in the early morning.

After waking to the alarm at 3:30am I looked out the window and was happy to see the moon in eclipse.  I started observing with low power and quickly framed both the moon and Uranus in the same FOV (54x 1.3 degrees).  I was surprised how close Uranus was to the moon, just a modest nudge of the scope away.  At 170x Uranus was a bluish green disk, almost orb shaped, clearly not stellar, having a kind of heft to it.

Seeing and transparency were good indeed.  I quickly ramped up magnification on the moon to 340x and 554x, and the view was steady.  However magnification did not improve detail on the moon; you really need sunlight for contrast.  The moon's surface was blotched red.  I could observe the mountains along the moon's limb very well, which I had not studied closely before –clefts, ridgelines,

(continued on page 4)

(Lunar eclipse continued from page 3)
and peaks. They must be of gigantic proportions.
Later, as the moon started to come out of eclipse, I followed the edge of the earth's shadow as it passed over the moon. The shadowed part of the moon appeared smoky and flat, while the more lighted shadow edge revealed ghostly detail. I struggle for words to describe it. As the edge of the shadow passed over the crater Copernicus, for example, with its many wisps and tendrils of ejecta streaks, it looked in the eyepiece at 340x like a dramatic sunset over the Pacific – I felt I was looking at the earth's sky rather than the moon.
With the moon at maximum eclipse, and with the conditions as good as they were, I decided to check out some other objects. I could see fuzziness around the stars of Pleiades (M45)--which I initially thought was because my eyepiece had fogged up, but seeing other stars sharp and clear confirmed I was really seeing nebulosity. The Orion Nebula (M42 & M43) at 170x filled the 0.6 deg. TFOV, with wonderful swirls and dark lanes in the nebula, and—adding to the thrill—I could easily see stars A-F in the Trapezium! This was my first time seeing six stars in this famous grouping. I made an attempt at the Horsehead Nebula, using an H-Beta filter and 35mm eyepiece to yield the supposedly ideal 5mm exit pupil, but no luck – likely observer error in either positioning the scope or just not having an eye trained to see it. I will try it again on the next superior night.
This was a wonderful experience and well worth the sleep deprivation and what surely will be a long day at work ahead.

# The Leonids Meteor Shower

The Leonids will peak on November 16 and November 17 in 2014. A waning crescent moon means that the sky will be dark enough to easily view the shower. We suggest that observers try their luck in the early morning of November 17.

The Leonids peak around mid-November. The Leonids can be seen by Northern and Southern Hemisphere observers.©iStockphoto.com
The Leonid meteor shower is annually active in the month of November and usually peaks around November 16. The shower is called Leonids because it's radiant or the point in the sky where the meteors seem to emerge from, lies in the constellation Leo.
The Leonids occur when the Earth passes through the debris left by the comet Tempel-Tuttle. The comet takes around 33 years to make one orbit around the Sun.
People can view about 20 meteors an hour at the peak of the Leonids meteor shower. While it is not necessary to look in a particular direction to enjoy a meteor shower – just lay down on the ground and look directly above and you are bound to see some meteors. Astronomers suggest lying down on the ground and looking at the sky between the East and the point right above you to view the Leonids. The best time to view the Leonids is just after midnight and right before dusk.

Credit: timeanddate.com

# All Three NASA Mars Orbiters Healthy After Comet Flyby October 19, 2014

*This artist's concept shows NASA's Mars orbiters lining up behind the Red Planet for their "duck and cover" maneuver to shield them from comet dust that may result from the close flyby of comet Siding Spring (C/2013 A1) on Oct. 19, 2014. Image Credit: NASA/JPL-Caltech*

All three NASA orbiters around Mars confirmed their healthy status Sunday after each took shelter behind Mars during a period of risk from dust released by a passing comet.
Mars Odyssey, Mars Reconnaissance Orbiter and the Mars Atmosphere and Volatile Evolution (MAVEN) orbiter all are part of a campaign to study comet C/2013 A1 Siding Spring and possible effects on the Martian atmosphere from gases and dust released by the comet. The comet sped past Mars today much closer than any other known comet flyby of Mars or Earth. Credit: NASA

Mangalyaan – India's Mars Orbiter Mission
The Indian Space Agency, ISRO had repositioned Mangalyaan to avoid any collision with the comet. At its closet, Siding Spring was 139,500 kilometers from Mars -- less than half the distance between Earth and moon.
"Signal confirming closest approach has just been received," the European Space Agency said on Twitter. Before the comet passed, it could be seen in space racing toward the bright planet, trailed by a tail of debris.
The ball of ice, dust and rocks is believed to have originated billions of years ago in the Oort Cloud, a distant region of space at the outskirts of the solar system.
The comet is around a mile wide and is only about as solid as a pile of talcum powder.

Credit: ISRO

## Kid Spot Jokes:

- **Why did the sun go to school?**
  (To get brighter)

- **Why did the cow go on the spaceship?**
  (To see the moooooooooon)

## Kid Spot Quiz:

1. **The Pleiades star cluster is also known by which other names?**
2. **The OBAFGKM system is used to classify what?**

## Kid Spot Night Sky Challenge: November 2014

⇒ Leonid Meteor Shower - Nov 17
⇒ Jupiter and Moon – Dawn
⇒ Mercury – Early morning
⇒ The Orion Nebula

http://skyandtelescope.com/observing/ataglance

*Image: M42 Orion Nebula*
*Photo Credit: Piller*

Kids, Fall back!   Daylight Savings Time Ends Nov 2, 2014.  Set clocks back one hour at 2 am and sleep in an extra hour!

## Constellations

### Pegasus

is a constellation in the northern sky, named after the winged horse Pegasus in Greek mythology. It was one of the 48 constellations listed by the 2nd-century astronomer Ptolemy, and remains one of the 88 modern constellations.


With an apparent magnitude varying between 2.37 and 2.45, the brightest star in Pegasus is the orange supergiant Epsilon Pegasi, also known as Enif, which marks the horse's muzzle.  Alpha (Markab), Beta (Scheat), and Gamma (Algenib), together with Alpha Andromedae (Alpheratz or Sirrah) form the large asterism known as the Square of Pegasus.

Pegasus is bordered by Andromeda to the north and east, Lacerta to the north, Cygnus to the northwest, Vulpecula, Delphinus and Equuleus to the west, Aquarius to the south and Pisces to the south and east.

The three-letter abbreviation for the constellation, as adopted by the IAU in 1922, is 'Peg'.

Deep Sky Object M15 (NGC 7078), located close to Enif, is a globular cluster of magnitude 6.4, 34,000 light-years from Earth. It is a Shapley class IV cluster, which means that it is fairly rich and concentrated towards its center.

M15 was discovered by Jean-Dominique Maraldi in 1746 and included in Charles Messier's catalogue of comet-like objects in 1764. At an estimated 12.0 billion years old, it is one of the oldest known globular clusters.

Credit:
Wikipedia

**Kid Spot Quiz Answers:**

**1) The Seven Sisters or M45.**
**2) The surface temperature and color of the stars.**

## SJAA Library!

From Sukhada Palav

SJAA offers another wonderful resource, a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers!
Also, we are always looking to add more fun and educational resources to our library. I have researched and picked a few good kid friendly books that would be an excellent addition our SJAA library! Please check out our wish list on the SJAA webpage:

http://www.sjaa.net/sjaa-library/

## Telescope Fix It Session

*Pictured above: Phil Chambers, Ed Wong and Dave Ittner busy at work*

**Fix It Day**, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. The Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

Weather doesn't slow this program down, so even if it's windy, cold and rainy outside, Fix It day goes on. Big thanks to go Ed Wong, Phil Chambers, and Dave Ittner for being the gear experts who faithfully make themselves available at Fix It day!

## School Star Parties

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill. For those who are outside of that area, please see our suggestions on the SJAA webpage under 'Classes-Programs / School Star Party'
The co-coordinator for SJAA is Jim Van Nuland.

Please see the SJAA webpage for a step-by-step procedure for setting up a school star party.

# From the Board of Directors

## Announcements

### SJAA Reaches 60 Years

SJAA has hit the 60 year old mark and is still going strong! Keep an eye out for announcements regarding commemorating the occasion.

### Advanced Loaner Program (ALP)

Dave Ittner is looking for someone to take over ALP by end of calendar year. If interested please contact Dave Ittner or any board member for more information.

### Imaging SIG Status

The board announced that Mark Striebeck is the new chair of the Imaging Group.

### Swap Meet (Sunday Nov 9)

The Annual swap meet is scheduled for Sunday November 9th from 11am-4pm.

## Board Meeting Excerpts October 11, 2014

### In attendance

Rob Jaworski, , Lee Hoglan, Ed Wong, Dave Ittner, Michael Packer, Rich Neuschaefer, Bill O'Neal, Teruo Utsumi

### RCDO Incident

Ed reported imagers crashed Binocular Viewing at RCDO. They were disruptive and uncooperative. This presents a potential future problem. OSA has been contacted and asked to review their procedures. No issues reported yet at Mendoza Ranch.

### City of San Jose Facility Reuse Mtg

Rob attended a recent meeting at City Hall. CoSJ plans to re-qualify "service providers" such as SJAA. They will send RFQs (request for qualification) out second quarter 2015.

### Divesting Scopes at the Swap

Dave Ittner plans to sell current inventory at the annual swap meet per his earlier board message.

### Income Statement

Michael presented financial statements, including current status of accounts and income and expenses.

### Solar Eclipse Event

Michael acquired material for funnel projectors for the October 23 eclipse and will assemble at Houge prior to the event. *(Editor's note: See feature article from Rob Jaworski on page1; the SJAA eclipse observation event was a huge success).*

### Renewing Mendoza Ranch

Ed Wong reported that renewal paperwork for the annual permit has been submitted and he is waiting for a response from Mendoza Ranch.

### Beginners Class

Teruo met with Marianne Damon. Had good exchange of ideas. Teruo will organize a meeting w/ Rob, Bill and others who are interested.

SJAA Contacts

| | |
|---|---|
| President: | Rob Jaworski |
| Vice President: | Lee Hoglan |
| Treasurer: | Michael Packer |
| Secretary: | Teruo Utsumi |
| Director: | Greg Claytor |
| Director: | Dave Ittner |
| Director: | Ed Wong |
| Director: | Rich Neuschaefer |
| Director: | Bill O'Neil |
| Beginner Class: | pending |
| Fix-it Program: | Ed Wong |
| Imaging SIG: | Mark Striebeck |
| Library: | Sukhada Palav |
| Loaner Program: | Dave Ittner |
| Ephemeris Newsletter - | |
| Editor: | Sandy Mohan |
| Prod. Editor: | Tom Piller |
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| Questions: | Lee Hoglan |
| Quick STARt | Dave Ittner |
| Solar: | Michael Packer |
| School Events: | Jim Van Nuland |
| Speakers: | Teruo Utsumi |

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Articles for publication should be submitted by the 20th of the previous month.

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# San Jose Astronomical Association Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

☐ **New** ☐ **Renewal** (Name only if no corrections)

**Membership Type:**

☐ Regular — $20

☐ Regular with Sky & Telescope — $53

☐ Junior (under 18) — $10

☐ Junior with Sky & Telescope — $43

Subscribing to Sky & Telescope magazine through the SJAA saves you $5 off the regular rate. (S&T will not accept multi-year subscriptions through the club program. Allow 2-3 months lead time.)

☐ **I prefer to get the Ephemeris newsletter in print form (Add $10 to the dues listed on the left). The newsletter is always available online at:**

**http://www.sjaa.net/sjaa-newsletter-ephemeris/**

Questions? Send e-mail to
**sjaamemberships@gmail.com**

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to “SJAA”, or join/renew at **http://www.sjaa.net/join-the-sjaa/**

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**Address:** ____________________

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